DO NOW — RECALL

Synoptic Revision 2 — Final Test Preparation

Name: _____ Class: _____ Date: _____

1. A username must contain at least one character. Give an example of invalid input and explain what the program should do.

2. For a full adder with A = 1, B = 0 and CarryIn = 1, calculate Sum and CarryOut.

3. A loop starts with total = 0 and adds the values 2, 4 and 6 once each. What is total after the third iteration?

4. A half adder has inputs A and B. Its Sum output is A XOR B and Carry output is A AND B. For inputs (A,B) = (0,1), (1,0) and (1,1), calculate both outputs and explain why the final case produces a carry.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

An empty string is invalid because it contains no characters. The program should reject it, display an appropriate message and not store the username.

2. Model answer

$1 + 0 + 1 = 2 = 10_2$, so Sum = 0 and CarryOut = 1.

3. Model answer

12. After each iteration the totals are 2, 6 and 12.

4. Model answer

(0,1) → Sum 1, Carry 0; (1,0) → Sum 1, Carry 0; (1,1) → Sum 0, Carry 1. In the last case $1 + 1 = 10_2$, so the sum bit is 0 and the carry bit is 1.

By the end, you can…

• complete one focused OCR block from each part of 1.4;

• mark immediately and identify why any mark was lost;

• retry weak questions without looking at the answer.

Vocabulary — know these terms

Recall the term — normalised

Recall the term — pointer

Recall the term — LIFO

Recall the term — FIFO

Recall the term — hash

Recall the term — De Morgan

Recall the term — Karnaugh map

Recall the term — carry

NEW LEARNING

Read, annotate and explain

Attempt → mark → fix → retry

1.4.1 Data Types
1.4.2 Data Structures
1.4.3 Boolean Algebra

Spend most time on the section you rated red/amber last lesson.

Do not copy corrections

The retry proves the correction has stuck.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Use the mark value

Direction

right shift

Amount

two places

Final answer

Shift right by two places.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY: Bit shifts

00001100 is shifted two places left.

```
result = ?
effect = ?
```

Method / working:

Guided 2 — TRY: Binary search tree

Why can a BST be preferable to an unsorted array for searching?

```
"It is quicker."
```

Method / working:

Guided 3 — TRY: Boolean simplification

Simplify the expression.

```
¬(¬A ∧ ¬B) = ?
```

Method / working:

Guided 4 — TRY: Full adder

What does the circuit add?

```
It adds ...
```

Method / working:

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Data Types — OCR practice
Complete the Teams questions on negative binary, floating point and bit shifts. Mark them immediately.

Question 2

Data Structures — OCR practice
Complete the Teams questions on linked lists, BSTs and hash tables. Focus on precise explanation.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

Boolean Algebra — OCR practice
Complete the Teams questions on Boolean simplification, truth tables, K-maps and adders.

Question 4

Personal checklist
Write the two exact things you will check before handing in the test next lesson.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

Can you name the two exact things you will check before handing in the test?

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

Complete both parts
CODE:
result = 00110000
effect = multiply by 4

Guided 2 — CHECK

Improve the answer
CODE:
"It is quicker."
CODE:
A BST can reduce the remaining search space after each comparison, whereas an unsorted array may require a linear search.

Guided 3 — CHECK

Complete the same expression
CODE:
¬(¬A ∧ ¬B) = ?
CODE:
¬(¬A ∧ ¬B) = A ∨ B

Guided 4 — CHECK

Improve the answer
CODE:
It adds ...
CODE:
It adds A, B and Cin. S is the result bit and Cout is the carry-out.